

Chapter 2: Angles

Drill practice and mixed revision of all 11 lessons · 103 questions

NSW syllabus outcome

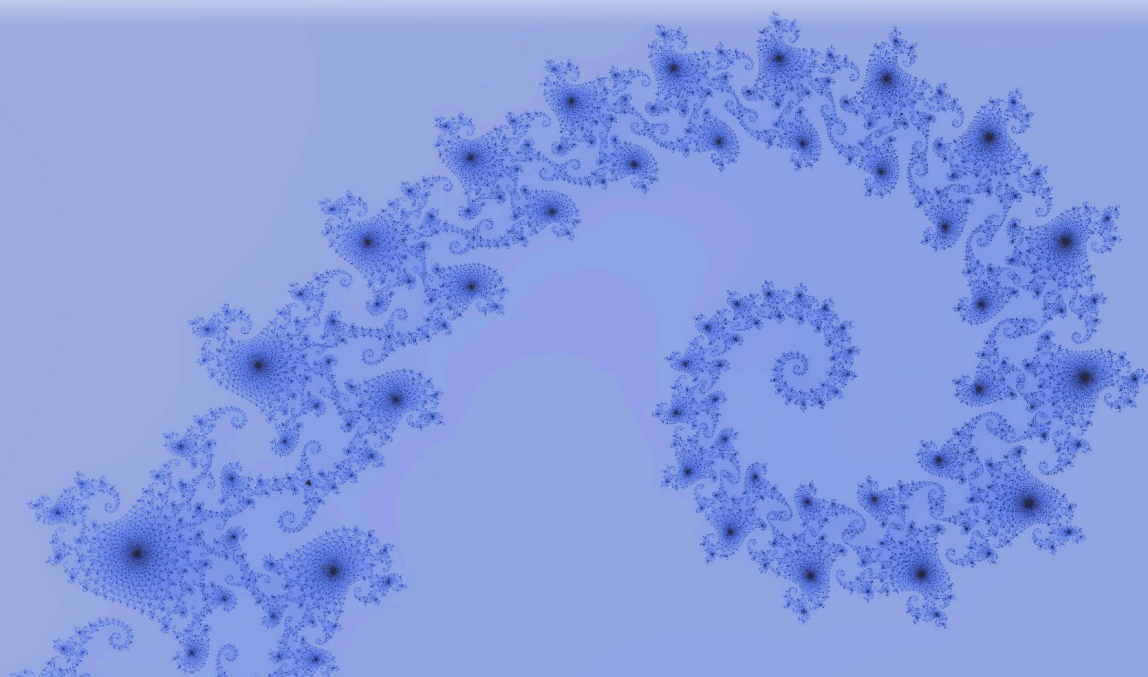
Stage 4 Measurement and space: MA4-ANG-C-01 applies angle relationships to solve problems, including those related to transversals on sets of parallel lines.

NAME

CLASS

CHAPTER

2



Homework. Part 1 is **drill practice**: quick questions on the main skill of each lesson in Chapter 2. Write each answer on its line. Part 2 is **mixed revision**, from **Easy** to **Challenging**. Each question is tagged with its lesson, so you can look back at that lesson's summary if you get stuck.

YOU DO On your own: Part 1, drill practice.

QUESTIONS 1–39

2.01 Naming angles**Easy** Which letter is the vertex of each angle?

① $\angle ZMT$

② $\angle BNW$

③ $\angle NYK$

④ $\angle XSW$

Medium Write the two arms of each angle.

⑤ $\angle PFD$

⑥ $\angle KPC$

⑦ $\angle HFK$

⑧ $\angle SFM$

2.02 Measuring and drawing angles**Easy** Reflex angles: write the angle you draw with the protractor ($360^\circ - \text{the angle}$).

⑨ 265°

⑩ 305°

⑪ 320°

⑫ 240°

Medium Draw each angle in your book, then label its type here.

⑬ 100°

⑭ 30°

⑮ 195°

⑯ 270°

2.03 Classifying angles**Easy** Classify each angle: acute, right, obtuse, straight, reflex or revolution.

⑰ 125°

⑱ 315°

⑲ 190°

⑳ 40°

Medium Add the two angles, then classify the total.

㉑ $115^\circ + 140^\circ$

㉒ $190^\circ + 65^\circ$

㉓ $60^\circ + 120^\circ$

㉔ $135^\circ + 15^\circ$

2.04 Complementary and supplementary angles**Easy** Find the complement (they add to 90°).

㉕ 67°

㉖ 28°

㉗ 16°

㉘ 1°

Medium Are the two angles complementary (C), supplementary (S) or neither (N)?

㉙ 65° and 25°

㉚ 100° and 38°

㉛ 57° and 33°

㉜ 92° and 129°

2.05 Angles at a point and vertically opposite angles**Easy** Angles on a straight line add to 180° . Find a.

㉝ 154° and a°

㉞ 54° and a°

㉟ 144° and a°

㊱ 169° and a°

Medium Two lines cross. One angle is given. Find the vertically opposite angle (v) and the angle next to it (n).

㊲ 117°

㊳ 50°

㊴ 19°

YOU DO

On your own: Part 1, drill practice.

QUESTIONS 40–69

2.06 Parallel and perpendicular lines

Easy Write the symbol for each statement: $\parallel$ (parallel) or $\perp$ (perpendicular).

40 EF meets GH at 90°

41 the rails of a train track

Medium True or false?

42 Two lines perpendicular to the same line are parallel.

43 Lines that meet at 89° are perpendicular.

2.07 Corresponding angles

Easy The lines are parallel. Corresponding angles are equal. Find the corresponding angle.

44 45°

45 37°

46 79°

47 91°

Medium The lines are parallel. Find a , then the angle next to it on the line (b).

48 corresponding to 100°

49 corresponding to 120°

50 corresponding to 130°

2.08 Alternate angles

Easy The lines are parallel. Alternate angles are equal. Find the alternate angle.

51 48°

52 31°

53 41°

54 91°

Medium The lines are parallel. Find the alternate angle (a) and the co-interior angle (c).

55 134°

56 107°

57 154°

2.09 Co-interior angles

Easy The lines are parallel. Co-interior angles add to 180° . Find the co-interior angle.

58 147°

59 140°

60 145°

61 157°

Medium Co-interior angles on parallel lines. Find y .

62 $y^\circ + y^\circ + 158^\circ = 180^\circ$

63 $y^\circ + y^\circ + 142^\circ = 180^\circ$

64 $y^\circ + y^\circ + 150^\circ = 180^\circ$

2.10 Finding unknown angles on parallel lines

Easy Which rule? Write corresponding, alternate or co-interior.

65 upside-down F shape

66 backwards F shape

Medium The lines are parallel. One angle is given. Find the other three angles at the same crossing.

67 33°

68 93°

69 54°

YOU DO

On your own: Part 1, drill practice.

QUESTIONS 70–75

2.11 Proving lines are parallel

Easy Is $AB \parallel CD$? Write yes or no.

70 Alternate: 132° and 132°

71 Co-interior: 76° and 104°

72 Alternate: 64° and 64°

Medium Find x so that $AB \parallel CD$.

73 co-interior angles: 55° and x°

74 co-interior angles: 75° and x°

75 co-interior angles: 150° and x°

YOU DO

On your own: Part 2, mixed revision.

QUESTIONS 76–85

76

Easy

Which letter is the vertex?

$\angle LMN$ Lesson 2.01

78

Easy

True or false.

Railway tracks are parallel. Lesson 2.06

80

Easy

Is $AB \parallel CD$?

Co-interior: 105° and 85° Lesson 2.11

82

Easy

Find the unknown angle.

Point: 95° , 145° and g° Lesson 2.05

84

Easy

Which rule is this?

Z shape Lesson 2.10

77

Easy

The lines are parallel. Find the co-interior angle.

132° Lesson 2.09

79

Easy

Classify the angle.

154° Lesson 2.03

81

Easy

The lines are parallel. Find the alternate angle.

A: 37° Lesson 2.08

83

Easy

What angle do you draw with the protractor to make this reflex angle?

225° Lesson 2.02

85

Easy

The lines are parallel. Find the corresponding angle.

84° Lesson 2.07

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 86–95

86 **Easy**

Find the supplement.

S of 128° Lesson 2.04

88

Medium What do co-interior angles add to on parallel lines? Lesson 2.09

90

Medium Classify 200° and 180° . Lesson 2.03

92

Medium What letter shape do alternate angles make? Lesson 2.08

94

Medium Draw $\angle ABC = 135^\circ$. Lesson 2.02

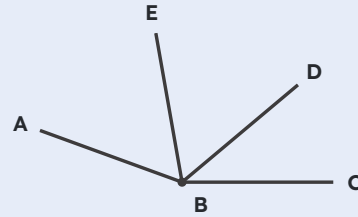
87

Medium

Use the diagram.

Name the angle between arms BC and BE.

Lesson 2.01



89

Medium Name 2 perpendicular lines in your classroom. Lesson 2.06

91

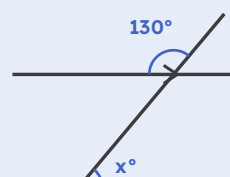
Medium Co-interior angles are 105° and 85° . Are the lines parallel? Explain. Lesson 2.11

93

Medium Two lines cross. One angle is 38° . Find the vertically opposite angle. Lesson 2.05

95

Medium Find all the variables. Give reasons. Lesson 2.10



YOU DO On your own: Part 2, mixed revision.

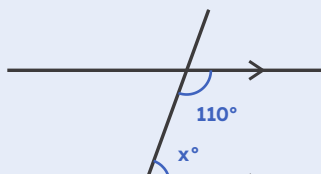
QUESTIONS 96–103

- 96** **Medium** What shape do corresponding angles make? Lesson 2.07

- 98** **Challenging** Draw and label two adjacent angles $\angle ABC$ and $\angle CBD$. Lesson 2.01

- 100** **Challenging** Find each variable. Give a reason.
Two lines cross. One angle is 65° . Find the other three angles. Lesson 2.05

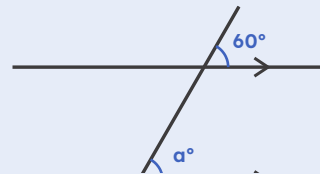
- 102** **Challenging** Co-interior angles are $3y^\circ$ and $6y^\circ$. Find y . Lesson 2.09



- 97** **Medium** Find the supplement of 118° . Lesson 2.04

- 99** **Challenging** Think it through.
How many reflex angles are there between 200° and 220° (whole degrees)? Lesson 2.03

- 101** **Challenging** Corresponding angles are $(2x + 20)^\circ$ and 60° . Find x . Lesson 2.07



- 103** **Challenging** Write a proof.
Draw a diagram where the lines are NOT parallel and explain how you know. Lesson 2.11

YOU DO

On your own: extra working space. Write the question number next to your working.